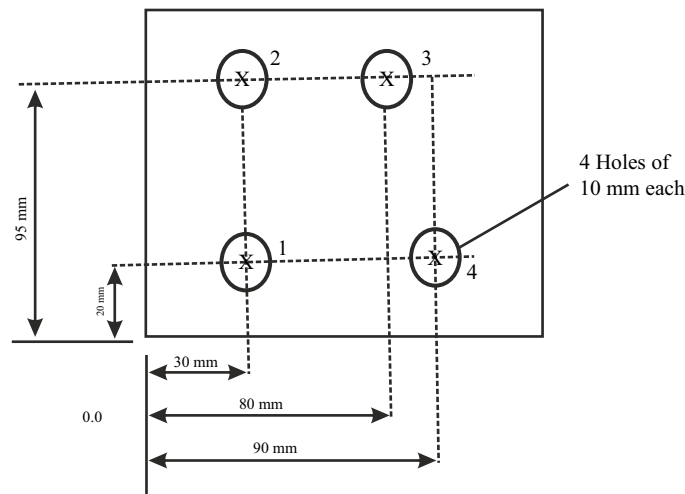


- Q.32 What do you mean by manual part programming ? Explain.
- Q.33 Explain any one canned cycle with diagram.
- Q.34 Define the following
 a) Encoders b) MCU
- Q.35 What are the areas of applications, where CNCs is profitable?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is NC words? Explain all the NC words.
- Q.37 Write a short note on:
 a) FMS systems b) Group Technology
 c) Concept of A.I.
- Q.38 Write a part program for drilling of 4 holes as shown in Fig. Table Spindle speed = 1440 R.P.M., and Feed Rate as 140 mm/min. The position of Drill Edge is Zero, 80 mm above the table surface. The thickness of plate is 25 mm.



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5th Sem / Mech, Mechatronics, Prod, T&D, CAD/CAM, Mech. Engg. (Fabrication Tech.). Mechanical Engg. (CAD/CAM Design & Robotics)

Subject:- CNC Machines and Automation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 ASRS stands for :-
 a) Automatic action system
 b) Automatic storage & Retrieval system
 c) Automatic asynchronous system
 d) None of the above
- Q.2 _____ in which the parts having similarities in geometry, manufacturing process and function.
 a) Group Technology b) AGVs
 c) FMS d) None of the above
- Q.3 Opto-interrupter work as a
 a) LVDT sensing device
 b) Tachometer
 c) Limit switches
 d) Any of the above
- Q.4 Absolute co-ordinate system is defined by
 a) Code G-90 b) Code G-91
 c) Code G-76 d) Code G-71

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- Q.5 _____ is a number of steps repeated over a number of equal steps for a previously fixed number of items.
- Macros
 - Parametric subroutines
 - Mirror image
 - Do-loops
- Q.6 G-79 code used for
- Basic mill cycle
 - Bore cycle
 - Dwell cycle
 - Peck drill cycle
- Q.7 _____ Is a device that detect/measure a signal or stimulus.
- Actuator
 - Transducer
 - CAD
 - Sensor
- Q.8 In robotics _____ joint provide translation sliding motion between input and output link
- Twisting
 - Orthogonal
 - Linear
 - Rotational
- Q.9 LVDT is a _____ positional transducer.
- Passive
 - Active
 - Rotational
 - None of the above
- Q.10 Expand DNC
- Double number control
 - Dedicated numeric center
 - Direct numerical control
 - Dimension-Inter control

SECTION-B

Note: Objective type questions. All questions are compulsory.
(10x1=10)

- Q.11 _____ logic is used in PLC.
- Q.12 $(78)_{10} = (?)_2$
- Q.13 A motor is used in control with feedback is _____
- Q.14 What is an axis?
- Q.15 CNC is a machine or system?
- Q.16 Define Canned cycle?
- Q.17 The basic robot elements are _____
- Q.18 Lathe bed is made of _____ material
- Q.19 When an Ac motor is used:
- Q.20 Expand AGV

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 Discuss advantages and disadvantages of CNC machine.
- Q.22 Explain pre-set and semi qualified tool with neat sketches.
- Q.23 Explain LVDT in brief.
- Q.24 Explain briefly about Flexible manufacturing system with their components.
- Q.25 Enlist the various Automation strategies.
- Q.26 Write the various problems related to Mechanical components.
- Q.27 What do you understand by swarf removal.
- Q.28 Explain Potentiometer for linear position with the help of diagram.
- Q.29 Write a short note on cutter radius compensation.
- Q.30 Differentiate between open loop and close loop system.
- Q.31 Describe pallets in CNC. Explain its different system / types.